

A priori estimates across the subcritical Lane–Emden hyperbola

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Abstract

We record the consequences of the subcritical entire Liouville theorem in OpenAI’s 2026 Hénon–Lane–Emden preprint for the unweighted system $-\Delta u = v^p$, $-\Delta v = u^q$. For $n \geq 3$, $p, q > 1$, and $1/(p+1) + 1/(q+1) > (n-2)/n$, the established reductions of Poláčik–Quittner–Souplet give universal pointwise estimates on arbitrary proper domains and zero-Dirichlet half-space nonexistence. We detail the resulting gradient and exterior estimates and the boundary blow-up argument giving uniform bounds and compactness on fixed bounded C^2 domains, with exact function spaces. The full subcritical scope is an immediate consequence of the cited Liouville input; no independent solution of that conjecture or new Liouville-to-estimate machinery is claimed.

1 Scope, provenance, and statement

Let

$$n \geq 3, \quad p, q > 1, \quad \frac{1}{p+1} + \frac{1}{q+1} > \frac{n-2}{n}, \quad \alpha = \frac{2(p+1)}{pq-1}, \quad \beta = \frac{2(q+1)}{pq-1}. \quad (1)$$

Throughout, a domain is a nonempty connected open subset of $\mathbb{R}^n$. A nonnegative classical solution means $u, v \in C^2(\Omega)$, $u, v \geq 0$, satisfying

$$-\Delta u = v^p, \quad -\Delta v = u^q \quad \text{pointwise in } \Omega. \quad (2)$$

Zero-Dirichlet solutions additionally belong to $C(\overline{\Omega})$ and vanish on $\partial\Omega$. These statements do not assert regularity for arbitrary weak or distributional solutions.

The entire Liouville theorem used here is the unweighted corollary of OpenAI [1], at the source version specified below. The passage from bounded entire nonexistence to the universal domain and half-space conclusions is already Theorems 4.2 and 4.3 of Poláčik–Quittner–Souplet [2] (PQS). The fixed-domain Dirichlet reduction is also established; see Quittner–Souplet [3, Theorem 4.1(ii) and Remark 4.1(ii)], whose remark credits the pure-power case to PQS. We include its blow-up argument to distinguish a boundary estimate from a uniform Dirichlet bound and to specify compactness.

Thus the contribution of this consequence note is the explicit full-range formulation and checked compatibility of the input, boundary argument, and function spaces. Dimensions three and four, and portions of higher dimensions, already had the needed Liouville results; see [2, 4, 5]. The newly available scope is the remaining portions of (1), inherited from [1], rather than a new estimate formula. We do not claim priority for the transfer method or to be first to observe its implication. The source preprint carries September 24, 2026 as its manuscript date; the verified public release

is October 6, 2026 [7]. Our priority audit found no earlier explicit full-range application package in that release, but absence from a search does not certify priority.

Theorem 1. *Assume (1). There is $C = C(n, p, q) > 0$ such that every nonnegative classical solution of (2) on every proper domain $\Omega \neq \mathbb{R}^n$ satisfies, with $d(x) = \text{dist}(x, \partial\Omega)$,*

$$\begin{aligned} u(x) &\leq Cd(x)^{-\alpha}, \quad |\nabla u(x)| \leq Cd(x)^{-\alpha-1}, \\ v(x) &\leq Cd(x)^{-\beta}, \quad |\nabla v(x)| \leq Cd(x)^{-\beta-1}. \end{aligned} \tag{3}$$

No boundary regularity or boundary condition is required. If $\Omega \supset \{x : |x| > R\}$ for some $R > 0$, then for $|x| \geq 2R$ the same estimates hold with $d(x)$ replaced by $|x|$ (after changing C).

Every nonnegative zero-Dirichlet classical solution on $H = \{x_n > 0\}$ in $C^2(H) \cap C(\overline{H})$ is zero, with no boundedness or growth assumption.

For each fixed bounded C^2 domain Ω , its nonnegative zero-Dirichlet classical solutions satisfy

$$\|u\|_{L^\infty(\Omega)} + \|v\|_{L^\infty(\Omega)} \leq C(\Omega, n, p, q). \tag{4}$$

For every $1 < r < \infty$ and $0 < \gamma < 1$, this solution set is bounded and sequentially compact in

$$[W^{2,r}(\Omega) \cap W_0^{1,r}(\Omega)]^2 \quad \text{and} \quad [C^{1,\gamma}(\overline{\Omega})]^2,$$

respectively, with the indicated norm topologies. If $\partial\Omega$ is $C^{2,\theta}$, $0 < \theta < 1$, there are uniform $C^{2,\theta}(\overline{\Omega})$ bounds, and sequential compactness in $[C^{2,\eta}(\overline{\Omega})]^2$ for every $0 < \eta < \theta$. Constants for these norm bounds may depend on the fixed domain and the indicated regularity exponent. No $C^2(\overline{\Omega})$ conclusion is asserted from C^2 boundary regularity alone.

2 Entire input and interior normalization

Lemma 2. *Under (1), the only bounded nonnegative classical solution of (2) on $\mathbb{R}^n$ is $(0, 0)$.*

Proof. The unweighted corollary of [1] excludes strictly positive $C^2(\mathbb{R}^n)$ solution pairs, even without boundedness. If either nonnegative component vanishes at a point, the strong maximum principle for its superharmonic equation makes it identically zero. Its equation then forces the other component to be zero. Otherwise both components are strictly positive, and the cited theorem applies. $\square$

The source input is substantial, not a hypothesis hidden in notation. Our dependency audit inspected its Newton-potential reductions, localized virial identity, interval comparison, and energy-relative layer-loss estimate before the scaling contradiction. The accompanying package gives the exact source hashes and a reconstruction of the unweighted argument. The actual Lean source declaration is separately identified in the audit; its formal-verification status is reported there. A complete pinned Lean kernel build was not reproduced, and this follow-on theorem is not claimed to be formalized. The mathematical dependency acceptance here rests on the analytic proof audit.

For clarity, we recall the normalization underlying PQS Theorem 4.3. The scaling identities are

$$p\beta = \alpha + 2, \quad q\alpha = \beta + 2, \quad U(y) = \lambda^\alpha u(x + \lambda y), \quad V(y) = \lambda^\beta v(x + \lambda y). \tag{5}$$

The pair (U, V) satisfies the same equations. Put $M = u^{1/\alpha} + v^{1/\beta}$. For a nonzero solution, the maximum principle ensures $M > 0$. If no universal bound for Md exists, choose solutions with $M(y_k)d(y_k) > 2k$. The doubling lemma [2, Lemma 5.1] selects x_k with

$$M(x_k) \geq M(y_k), \quad M(x_k)d(x_k) > 2k, \quad M(z) \leq 2M(x_k) \quad (|z - x_k| \leq k/M(x_k)).$$

One can see the selection directly: whenever the last property fails, move by at most k/M to a point with more than twice that value. The sum of movements is at most $2k/M(y_k)$, strictly less than $d(y_k)$; if the process never stops, local boundedness at its interior limit contradicts $M \rightarrow \infty$.

With $\lambda_k = M(x_k)^{-1}$, the rescaled domains contain B_k , $U_k^{1/\alpha}(0) + V_k^{1/\beta}(0) = 1$, and $U_k^{1/\alpha} + V_k^{1/\beta} \leq 2$ on B_k . Interior elliptic estimates first give local $W^{2,r}$ bounds, then $C^{1,\gamma}$ bounds for suitable $r > n$. Since $p, q > 1$, the powers are Lipschitz on bounded nonnegative ranges. Interior Schauder estimates therefore give uniform $C^{2,\gamma}$ bounds on smaller balls. A diagonal subsequence converges in $C_{\text{loc}}^2(\mathbb{R}^n)$ to a bounded nonnegative pair satisfying the equations and the same nonzero normalization. This contradicts Lemma 2. Consequently $Md \leq C$, which proves the pointwise part of (3) with constants depending only on n, p, q .

3 Gradients, exterior decay, and the half-space

Fix $x \in \Omega$ and $\rho = d(x)/2$. On $B_\rho(x)$ the distance is at least $d(x)/2$. Applying the pointwise bounds to the scaling (5) with $\lambda = \rho$ gives uniformly bounded U, V and right sides on B_1 . The interior gradient estimate for Poisson's equation yields

$$|\nabla U(0)| \leq C(\|U\|_{L^\infty(B_1)} + \|V^p\|_{L^\infty(B_1)}) \leq C,$$

and similarly for V . Rescaling proves the gradient part of (3). For the exterior conclusion, every boundary point lies in $\overline{B}_R$, so $d(x) \geq |x| - R \geq |x|/2$ when $|x| \geq 2R$.

PQS Theorem 4.2 assumes exactly the bounded nonnegative entire nonexistence of Lemma 2, with $p, q > 1$, and excludes every nontrivial nonnegative $C^2(H) \cap C(\overline{H})$ zero-Dirichlet solution, bounded or unbounded. This proves the asserted half-space conclusion. Its full conclusion is needed here; an arbitrary-domain distance estimate by itself diverges at the boundary and would not prove (4).

4 Fixed-domain boundary blow-up and compactness

We give the remaining argument explicitly. On a bounded C^2 domain, a classical zero-Dirichlet pair continuous on the closure already belongs to $W^{2,r}(\Omega) \cap W_0^{1,r}(\Omega)$ for every $1 < r < \infty$: its right sides are bounded; solve the corresponding linear Dirichlet problems, and compare by the maximum principle. Indeed the difference is harmonic, continuous on the closure, and zero on the boundary. The global Poisson estimates on a $C^{1,1}$ domain (which a bounded C^2 domain is) give the stated regularity. We use also their local boundary versions and Sobolev embeddings; standard forms are in [6].

Suppose (4) fails. For a sequence of solutions set

$$A_k = \max_{\overline{\Omega}}(u_k^{1/\alpha} + v_k^{1/\beta}) \rightarrow \infty, \quad M_k(x_k) = A_k, \quad \lambda_k = A_k^{-1}.$$

The maximizing points are interior. Scaling about x_k gives $0 \leq U_k, V_k \leq 1$ and $U_k^{1/\alpha}(0) + V_k^{1/\beta}(0) = 1$. If $d(x_k)/\lambda_k \rightarrow \infty$ along a subsequence, the interior argument just given produces a nontrivial bounded entire solution, a contradiction.

Otherwise pass to a subsequence with $t_k = d(x_k)/\lambda_k \rightarrow t \geq 0$. Then $d(x_k) \rightarrow 0$. Choose a nearest boundary point b_k and rotate its inward normal into the y_n direction. Uniform C^2 boundary charts exist by compactness of $\partial\Omega$. Locally they have graphs $x_n = \varphi_k(x')$, with $\varphi_k(0) = D\varphi_k(0) = 0$ and uniformly bounded second derivatives. After scaling at b_k , the graphs are

$$y_n = \psi_k(y') = \lambda_k^{-1} \varphi_k(\lambda_k y'),$$

so $\psi_k \rightarrow 0$ in C^2 on every fixed compact set; the point x_k becomes $(0, t_k)$. Pull back by $(y', y_n) \mapsto (y', y_n + \psi_k(y'))$. The resulting uniformly elliptic coefficients tend to those of the Laplacian, and the lower order coefficients tend to zero. The charts have uniform $C^{1,1}$ control, the solutions and right sides are uniformly bounded, and the flat-boundary trace is zero. Local boundary $W^{2,r}$ estimates therefore give uniform $C^{1,\gamma}$ bounds on smaller compact half-balls for any $0 < \gamma < 1$, choosing r sufficiently large.

A diagonal subsequence converges in C^1 on compact subsets of $\overline{H}$ and in C^2 on compact subsets of H to a bounded nonnegative solution (U, V) , continuous and zero on ∂H . The transformed equations converge in distributions to the Laplacian equations; interior regularity upgrades the limit to $C^2(H)$. Normalization passes at $(0, t)$. If $t = 0$, continuity and the zero trace contradict normalization. If $t > 0$, the limiting pair is nontrivial, contradicting the half-space theorem. Both cases exclude the assumed blow-up, proving (4).

The uniform L^∞ bound now gives uniform global $W^{2,r}$ bounds for every finite $r > 1$. Compact Sobolev embeddings, using a larger r if necessary, give a subsequence converging in $C^{1,\gamma}(\overline{\Omega})$ for any fixed $\gamma < 1$. The limiting pair retains nonnegativity and zero trace, and its equations hold in distributions and then classically in the interior. The right sides converge uniformly. For each finite $r > 1$, the global linear Dirichlet estimate applied to differences gives

$$\|u_k - u\|_{W^{2,r}} \leq C_r \|v_k^p - v^p\|_{L^r} \rightarrow 0, \quad \|v_k - v\|_{W^{2,r}} \leq C_r \|u_k^q - u^q\|_{L^r} \rightarrow 0.$$

Thus the compactness is strong, not just weak $W^{2,r}$ compactness. On a $C^{2,\theta}$ domain, the already uniform Lipschitz bounds make the right sides uniformly $C^{0,\theta}$; global Schauder estimates give uniform $C^{2,\theta}$ bounds and compactness in $C^{2,\eta}$, $\eta < \theta$. This completes Theorem 1.

5 Limits and verification disclosure

The strict hyperbola and $p, q > 1$ are used as stated. No critical or supercritical case, weighted boundary estimate, parabolic estimate, or weak-solution regularity assertion is included. The conditional asymptotically-power extension in PQS Theorem 7.3 is established work; we do not present it as a new extension here. Spatially varying coefficients are outside this note.

The upstream input is the public commit `adc7f124` (full identity in the dependency ledger); source hashes, analytic dependency checks, priority evidence, and complete-package adversarial reviews accompany this note. Automated checks and reviews are evidence about a proof, not a substitute for it. AI tools were used extensively in research, drafting, and verification. This preprint has not undergone conventional human peer review or refereeing at publication. The author alone is listed; the upstream AI-produced theorem is attributed to OpenAI using its manuscript-specific citation.

References

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- [2] P. Poláčik, P. Quittner, and Ph. Souplet, *Singularity and decay estimates in superlinear problems via Liouville-type theorems. Part I: Elliptic equations and systems*, Duke Math. J. **139** (2007), 555–579. [doi:10.1215/S0012-7094-07-13935-8](https://doi.org/10.1215/S0012-7094-07-13935-8). The theorem statements were checked in the [author-hosted preprint](#).

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